Student Activity: Making a Trumpet

Purpose: To wire and code the Arduino to use three buttons as inputs to control the outputs which are sound frequencies that will simulate musical notes.

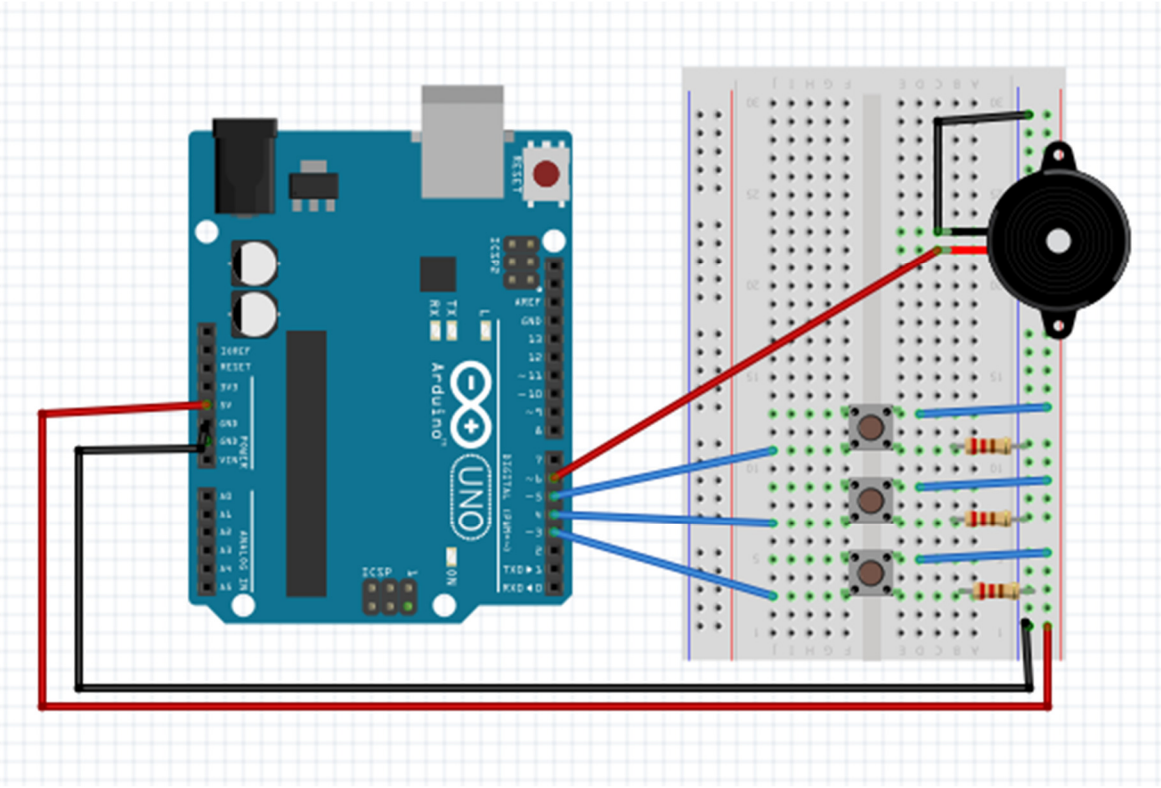
Materials: Computer, Arduino Uno Board, Piezo buzzer (speaker) USB cable, Breadboard, 3 - 220 Ω Resistors, 3 buttons, 10 Jumper Wires, and a potentiometer

New Components:

There are no new components for this activity.

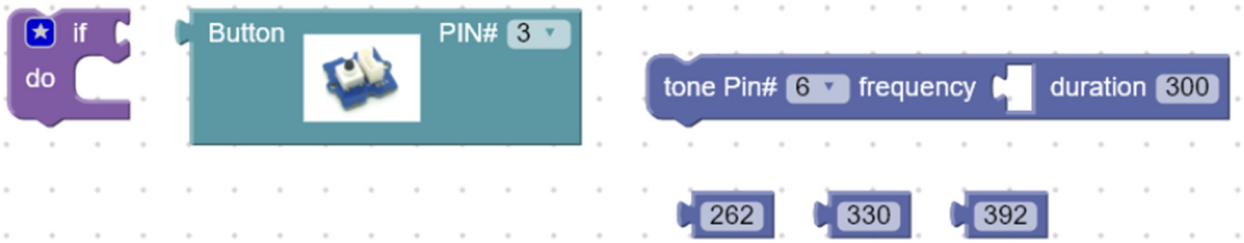
Procedure:

1. Wire the project using the following schematic. **HINT: WIRE ONE BUTTON AND THE SPEAKER. UPLOAD YOUR CODE AND SEE IF IT MAKES A TONE. WIRE A SECOND BUTTON AND SEE IF IT MAKES THE DESIGNATED TONE. IF THAT IS SUCCESSFUL, COMPLETE WIRING A THIRD BUTTON.**



2. You can create your code in the BlocklyDuino Editor then paste it into the Codebender website. Click BlocklyDuino Reference for a link to all the blockly codes.

3. Below are the blocks that you will need. Assign a button to Pins 3, 4, and 5. Your tone (speaker) will be assigned to pin 6. Assign the frequency 262 hz to pin 3, 330 hz to pin 4, and 392 hz to pin 5. The number blocks to input the frequencies are found in the math blocks or your blockly editor. If you are unsure of how to accomplish this or need some guidance watch the following video: **VIDEO: MAKING MUSIC**

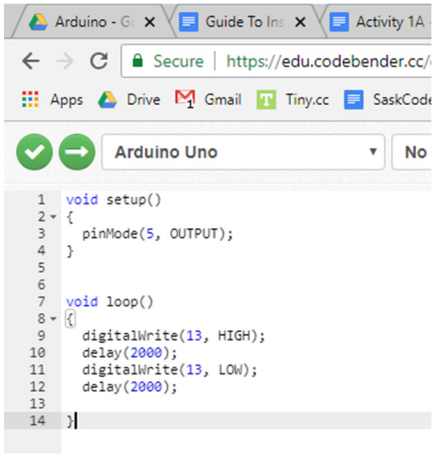


*****SAVE your sketch to your BlocklyDuino Folder that you created*****

4. Once you are done putting the blocks in the correct order, and have connected them, click on the Arduino button to convert the blocks into Arduino code.

5. Copy the Arduino Code from BlocklyDuino into Codebender. Hook up your arduino to your computer using the blue USB cable.





6. Upload your program to the arduino. Make sure the Arduino Uno board is hooked up using the supplied USB cord to your computer and click the green

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"upload" button.

7. Your program is now loaded. Observe your LED and complete your worksheet titled Assignment: Making Music.

NOTE: Keep your trumpet project as it will be needed for Activity 2C Trumpet with BLING.

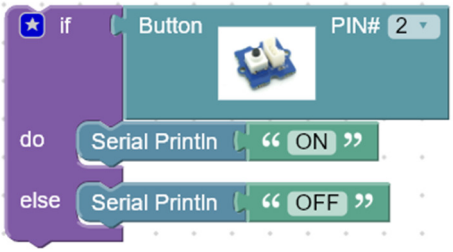
Troubleshooting: If your circuit is not working!

1. Double check your code! Did you put in every block? Are the settings correct?
2. Refresh the website and reload the code.
3. The next step is to check your circuit!
    a. Check your wiring! Is it the same as the circuit diagram? Does the circuit go from the positive numbered pin to the ground?
    b. Next check each component. Check the speaker, is it connected by wires to pin 6?
    c. Check the speaker. Rewire it to pin A0. Open a new Blocklyduino code and test the speaker by playing this code:



This should cause the speaker to chirp.

    d. Check the buttons one at a time. This can be achieved by connecting it and using the serial monitor built into Codebender. Open a new window of Blocklyduino and load this test program. Change the pin number to each of the button pins and test one at a time.



Open the serial monitor in Codebender and select "connect"As you hold the button down the serial monitor should change to on from off.

    e. Next check the individual wires in the circuit. This can be done by connecting an LED to pin 13 only then using the wire to ground the short pin. Do not connect the pin to the ground directly. (Shorting the wires) This will damage the UNO.
    f. For further information on using the serial monitor see Activity 4A Using a Serial

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Monitor Display.

Assignment Worksheet: Making a Trumpet

1. What happens when you press the buttons?

Extension Challenges:

2. Add another button to your Arduino Trumpet. Add the necessary code to make your fourth button work. You can find frequencies by researching them on the internet. Wire your button the pin#7. Code each button as follows:

Button 1 - C Button 2 - D Button 3 - E Button 4 - G

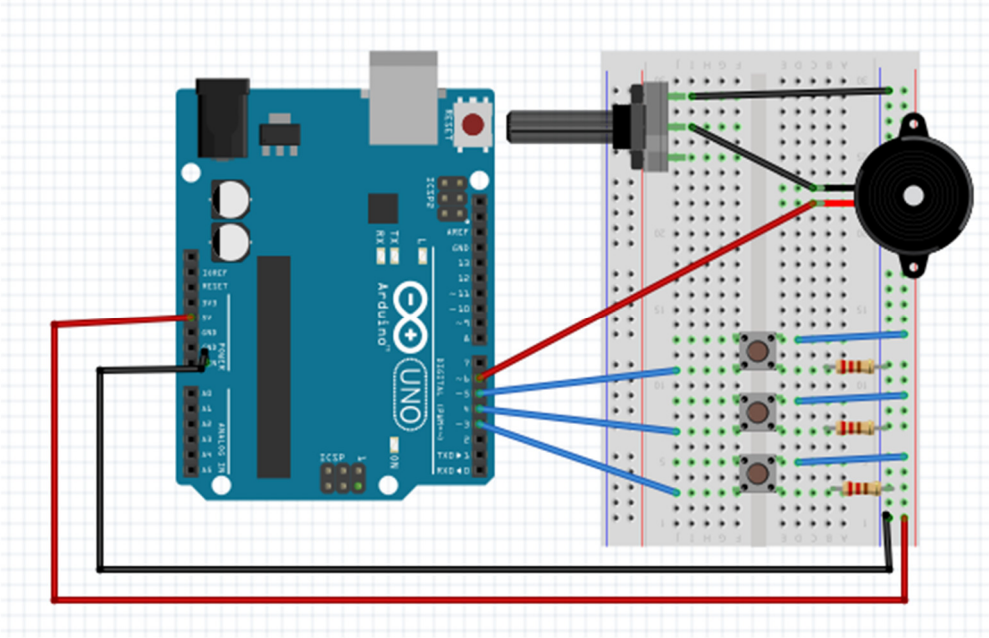
Find the music for "Mary had a Little Lamb" and play it for your teacher!

3. Add a potentiometer so that you can adjust the volume of your song. You will not have to change the code.

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